Q # 1 (Definitions)

1- **What** are the breadth and the depth of DSP?
2- **What** are the DSP education tasks?
3- **Define** the following items: Signal – System – Signal Processing. (State one example for each)
4- **Explain** with **drawing** the different classifications of the signal.
5- **Explain** the basic three operations performed in A/D converters.
6- **Find** the relation between the frequency of the discrete time signal, the frequency of the continuous time signals and the sampling frequency.
7- **State and explain** the sampling theorem.
8- In detail, **Explain** the quantization process used in A/D converter.
9- In detail, **Explain** the sampling process used in A/D converter.

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10- **Prove** that the DFT, X (k), of a sequence x (n) is periodic.

11- **Define** the following items: **(i)** Correlation operation. **(ii)** Correlation theorem. **(iii)** Correlation coefficient.
(iv) Convolution operation.

12- **Describe** the convolution operation and **deduce** its compact relations.

13- For LTIS **define** the following items: **(i)** Linearity. **(ii)** Causality. **(iii)** Stability. **(iv)** ROC. **(v)** Time Invariance. **(vi)** BIBO Condition. difference equation

14- **Write** the difference equations of both FIR and IIR filters.
15- **What** are the different steps of the digital filter design?
16- **Compare** between both filter types and mention a five from the digital filter advantages.

Q # 2- True and False